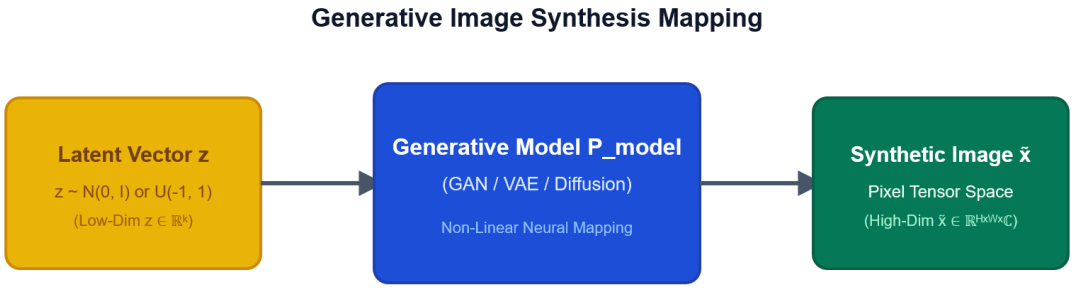
UNIT-II: Image Generation with Generative AI

1. Introduction to Image Generation

Image generation in Generative AI focuses on modelling the high-dimensional spatial distribution $P_{\text{data}}(X)$ of real-world images. The objective is to construct a generative model $P_{\text{model}}(X)$ capable of mapping low-dimensional noise vectors $z \in \mathbb{R}^k$ into realistic, synthetic multidimensional image tensors $\tilde{x} \in \mathbb{R}^{H \times W \times C}$.



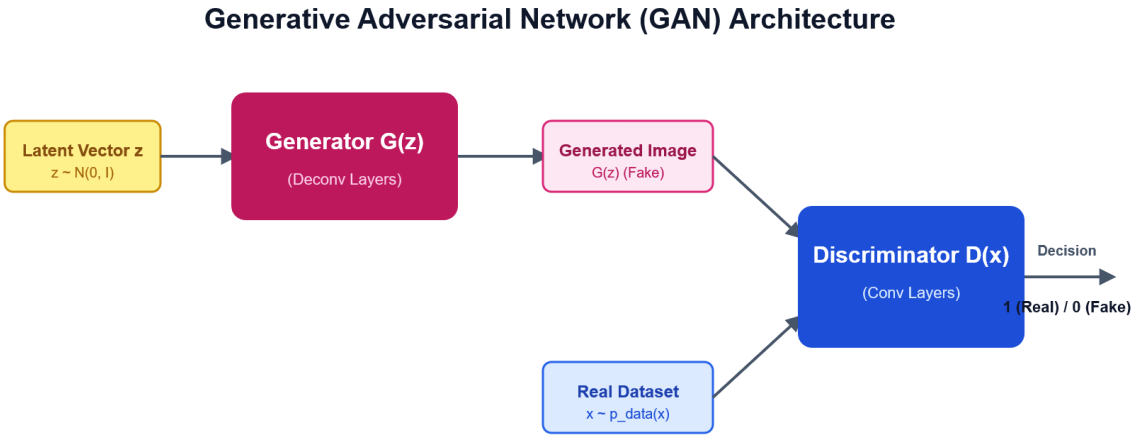
Core Image Modality Formulations

- **Spatial Tensors:** An image X is represented as a 3D tensor of shape (H, W, C) where H is height, W is width, and C is the color channel dimension ($C = 1$ for grayscale, $C = 3$ for RGB).
- **Latent Space Sampling:** Generative architectures sample a latent vector z from a prior distribution (typically isotropic standard normal distribution $z \sim \mathcal{N}(0, \mathbf{I})$ or uniform distribution $z \sim \mathcal{U}(-1,1)$) and project it into image space via neural networks.

2. Implementing Generative Adversarial Networks (GANs) for Image Generation

Generative Adversarial Networks (GANs), introduced by Ian Goodfellow et al. (2014), formulate image generation as a **two-player zero-sum minimax game** between two competing neural networks:

1. **Generator (G_θ):** Captures the data distribution and generates synthetic images $G(z)$ to fool the Discriminator.
2. **Discriminator (D_ϕ):** A binary classification network that estimates the probability that a given image came from the real training data $p_{\text{data}}(x)$ rather than from G.

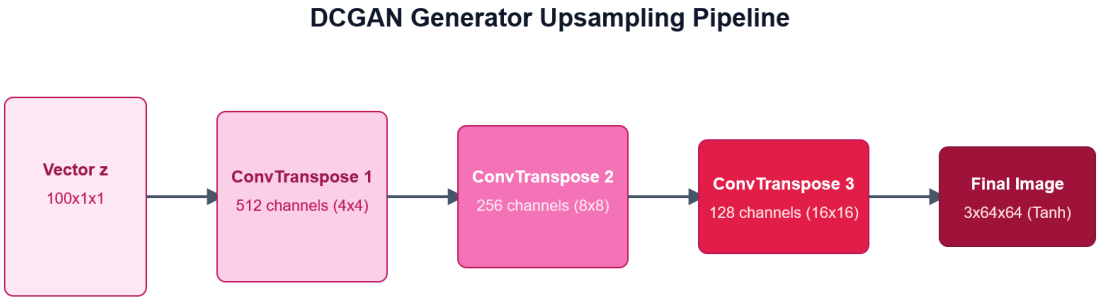


A. The Minimax Objective Function

The adversarial optimization problem is defined by the value function $V(D, G)$:

$$\min_G \max_D V(D,G) = \mathbb{E}_{x\sim p_{\text{data}}(x)}[\log D(x)] + \mathbb{E}_{z\sim p_z(z)}\left[\log\left(1 - D\left(G(z)\right)\right)\right]$$

- **Discriminator Maximization:** D tries to maximize $V(D,G)$ by assigning $D(x) \to 1$ for real images and $D\left(G(z)\right) \to 0$ for synthetic fake images.
- **Generator Minimization:** G tries to minimize $V(D,G)$ by forcing $D\left(G(z)\right) \to 1$, convincing D that the synthetic images are genuine.



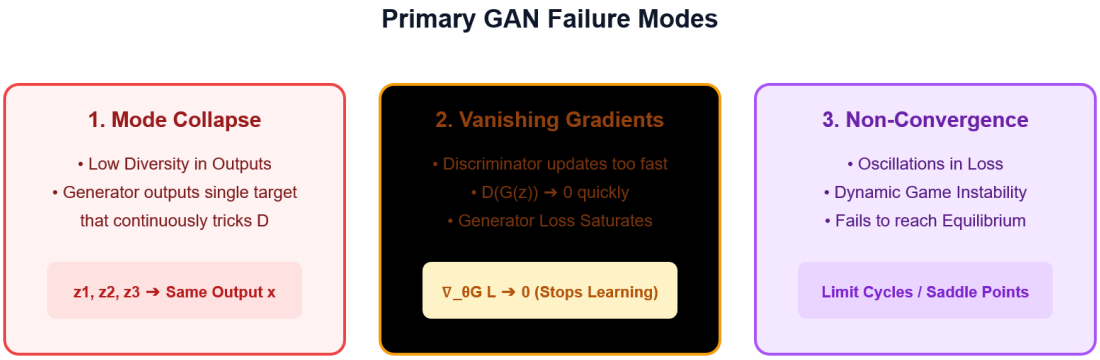
DCGAN Guidelines:

1. Replace spatial pooling layers with **strided convolutions** (Discriminator) and **fractionally-strided/transposed convolutions** (Generator).
2. Use **Batch Normalization** in both Generator and Discriminator to stabilize feature learning across channels.
3. Remove fully connected hidden layers for deeper architectures.
4. Use **LeakyReLU** activations in the Discriminator for all layers (0.2 slope).
5. Use **ReLU** activation in the Generator for all layers except the output layer, which uses tanh (scaling outputs to $[-1,1]$).

3. Training and Fine-Tuning GANs

Training GANs is notoriously unstable due to non-convergence issues caused by simultaneous optimization of non-convex loss surfaces.

A. Common GAN Training Failures



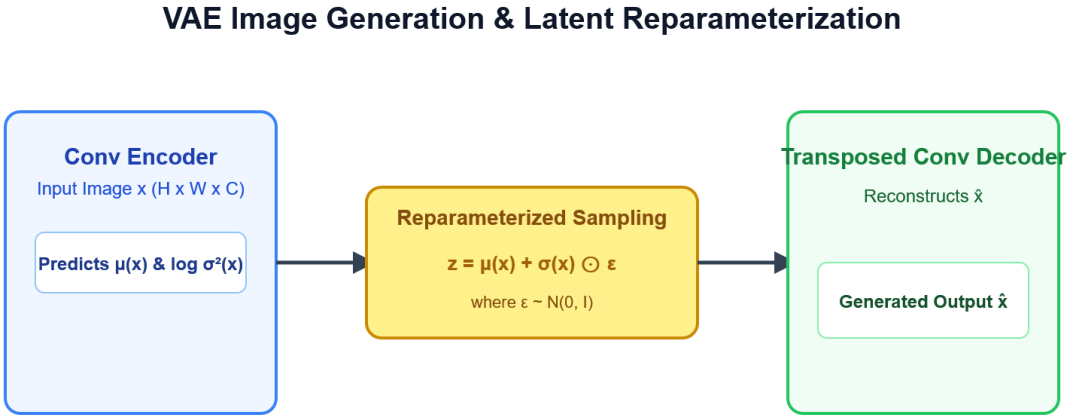
B. Stabilization & Fine-Tuning Techniques

- **Non-Saturating Heuristic Loss:** Generator maximizes $\log\left(D\left(G(z)\right)\right)$ instead of minimizing $\log\left(1 - D\left(G(z)\right)\right)$ to ensure strong gradients early in training.
- **One-Sided Label Smoothing:** Replace hard target labels (1.0 for real images) with soft targets (0.9) to prevent the Discriminator from becoming over-confident.
- **Two-Time-Scale Update Rule (TTUR):** Train Discriminator and Generator using different learning rates (typically $\alpha_D > \alpha_G$) to ensure D updates fast enough to provide meaningful feedback.

- **Spectral Normalization:** Normalizes the weight matrices W of each layer by their largest singular value (spectral norm $\sigma(W)$), constraining the network to be 1-Lipschitz continuous.

4. Generating Images with Variational Autoencoders (VAEs)

While GANs model image generation implicitly through adversarial games, **Variational Autoencoders (VAEs)** use explicit probabilistic modeling based on variational inference.



A. Mathematical Objective: ELBO for Images

The loss function minimizes negative Evidence Lower Bound ($\mathcal{L}_{\text{VAE}} = -\text{ELBO}$):

$$\mathcal{L}_{\text{VAE}}(\theta, \phi) = \mathcal{L}_{\text{reconstruction}}(x, \hat{x}) + D_{\text{KL}}\left(q_\phi(z|x) \,||\, p(z)\right)$$

- **Reconstruction Term:** Uses Mean Squared Error (MSE) or Binary Cross-Entropy (BCE) per pixel.
- **KL Divergence Penalty:** Forces latent space representations to form a continuous, smooth distribution without empty gaps:

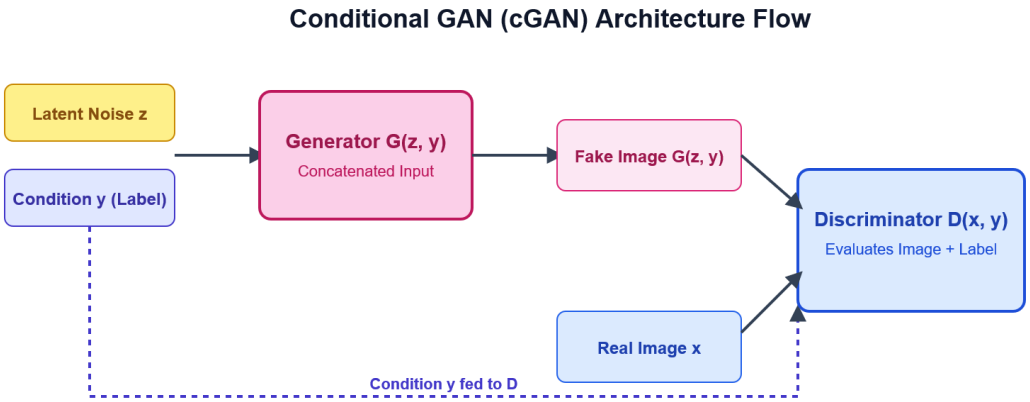
$$D_{\text{KL}}\left(\mathcal{N}(\mu, \sigma^2) \,||\, \mathcal{N}(0, \mathbf{I})\right) = -\frac{1}{2}\sum_{j=1}^{k}\left(1 + \log\left(\sigma_j^2\right) - \mu_j^2 - \sigma_j^2\right)$$

B. Trade-offs: GANs vs. VAEs in Image Generation

Attribute	Generative Adversarial Networks (GANs)	Variational Autoencoders (VAEs)
Output Image Quality	High spatial sharpness, rich fine details.	Slightly blurry images due to pixel-wise MSE/BCE averaging.
Training Stability	Unstable; subject to mode collapse & saturation.	Stable; optimizes a well-defined lower bound (ELBO).
Latent Space Structure	Discontinuous; random empty regions present.	Continuous & smooth; ideal for interpolation.
Likelihood Evaluation	Implicit (No direct probability estimation).	Explicit lower-bound estimation ($\log p\,(x)$).

5. Advanced Techniques in Image Generation

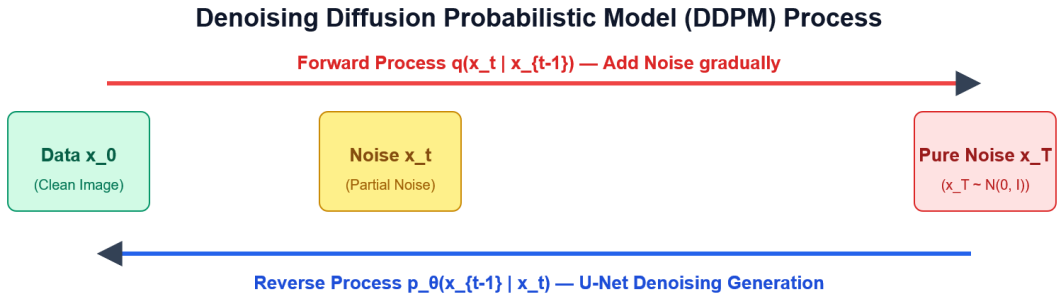
A. Conditional GAN (cGAN) Architecture



cGAN Objective Equation:

$$\min_G \max_D V(D,G) = \mathbb{E}_{x\sim p_{\text{data}}}[\log D\,(x|y)] + \mathbb{E}_{z\sim p_z}[\log(1 - D(G(z|y)|y))]$$

B. Denoising Diffusion Probabilistic Models (DDPM) Pipeline



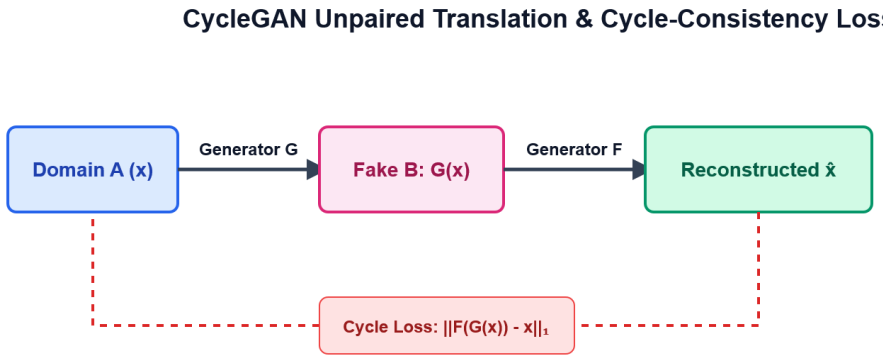
- **Forward Process:** Adds Gaussian noise over T timesteps according to a noise schedule β_t:

$$q(x_t|x_{t-1}) = \mathcal{N}\left(x_t; \sqrt{1-\beta_t}x_{t-1}, \beta_t \mathbf{I}\right)$$

- **Reverse Process:** Neural network $p_\theta(x_{t-1}|x_t)$ (typically a U-Net architecture) predicts the noise present at step t to reconstruct the clean image x_0.

6. Image and Video Generation Applications

CycleGAN Unpaired Image Translation Pipeline



Key Industry Applications:

1. **Super-Resolution (SRGAN / ESRGAN):** Upscales low-resolution images by predicting high-frequency details with perceptual and adversarial loss.
2. **Image-to-Image Translation (Pix2Pix & CycleGAN):**

- **Paired (Pix2Pix):** Maps input images to target domains (e.g., sketches → photo-realistic buildings).
- **Unpaired (CycleGAN):** Maps features across domains without paired examples using **Cycle-Consistency Loss**:

$$\mathcal{L}_{\text{cyc}}(G, F) = \mathbb{E}_x\left[||F(G(x)) - x||_1\right] + \mathbb{E}_y\left[||G(F(y)) - y||_1\right]$$

3. **Medical Image Generation:** Synthesizes realistic MRI, CT, and X-ray images for training diagnostic networks under strict privacy regulations.
4. **Video Generation & Temporal Consistency:** Uses 3D Convolutions and Temporal Attention layers to synthesize video sequences while avoiding frame flicker.